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Evaluation and application of andrographolide against largemouth bass ranavirus

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ABSTRACT

It is well-established that Largemouth bass ranavirus (LMBRaV) induces viral ulcerative syndrome, a highly transmissible and lethal viral disease afflicting largemouth bass aquaculture. Given the urgent need for available prevention and control methods, this study evaluated the antiviral activities of fifteen naturally occurring compounds derived from herbal medicines against LMBRaV using an *in vitro* cell model. Andrographolide (Andro) was one of the substances that was evaluated and showed highly effective anti-LMBRaV activity of a 99.2 % maximal inhibition rate of at 20 mg/L dosage. Mechanistic investigation revealed Andro prevented viral entry and directly inactivated the virus. Ultrastructural observations and fluorescence staining further supported Andro's protective effects on EPC cells from LMBRaV invasion. *In vivo* experiments demonstrated Andro's efficacy in reducing mortality rates in LMBRaV-infected largemouth bass by 61.54 % and significantly lowering the viral load. Additionally, Andro upregulated antiviral genes such as *ifn-γ*, *irf-3*, and *mx*, indicating its immunomodulatory properties. Histopathological analysis revealed that Andro alleviated tissue damage of liver, spleen, and kidney in the infected fish. Andro also changed the microbiotic composition of gut by increasing the amounts of good bacteria and reducing amounts of bad bacteria. These findings collectively highlight Andro's substantial antiviral and immunomodulatory potential, suggesting its promising application as a natural compound for combating LMBRaV infections in largemouth bass aquaculture and potentially other aquatic animal diseases.

1. Introduction

Largemouth bass (*Micropterus salmoides*), as a native species of southeastern United States, were introduced to China in 1983. At present, it has quickly rose to prominence as a major freshwater aquaculture species, with annual production surpassing 800,000 tons. However, the intense pressure to meet increasing demands has led to the widespread adoption of high-density farming practices. This has resulted in a significant increase in disease prevalence, including largemouth bass ulcerative syndrome, which causes up to 60 % mortality and becomes a severe threat to sustainable development of this industry [1]. While initially prevalent in Europe and America, the disease's impact in China has been growing [2]. The causative agent of this disease is Largemouth bass ranavirus (LMBRaV), a type of icosahedral, single-molecule, linear, double-stranded DNA virus attached to *Ranavirus*. Its genome size is 170–200 kb, and its diameter ranges from 120 to 300 nm. The major

capsid protein (MCP) of LMBRaV makes up 40–45 % of its total protein, and is extremely conserved. Therefore, the MCP protein is commonly used to differentiate among various ranaviruses [3]. To date, no effective treatment exists for this devastating viral disease, underscoring the urgent need for antiviral strategies to combat the LMBRaV epidemic.

China possesses a rich reservoir of traditional Chinese herbal resources. The utilization of these herbs or their extracted natural compounds has been shown to offer numerous advantages, including minimal environmental impact, reduced risk of resistance development, and limited harm to human health in animal farming [4,5]. The ever-increasing research evidence demonstrates the significant antiviral activity of various herbal plants or their derived natural compounds against aquatic animal diseases. For instance, the hydroxycoumarin can effectively inhibit Infectious Hematopoietic Necrosis Virus (IHNV) replication in *epithelioma papulosum cyprini* (EPC) cells, reduced IHNV-induced apoptosis, activated IFN-related gene expression, and

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Table 1
Sequences of primer pairs.

Genes	Primer sequence (from 5' to 3')	Reference/Accession numbers
EPC β -actin-Fq	GCTATGTGGCTCTTGACTTCGA	XM_038712406.1
EPC β -actin-Rq	CCGTCAGGCAGCTCATAGCT	
Largemouth bass β -actin-Fq	CAGGATGCAGAAGGAGATCACA	XM_038695351.1
Largemouth bass β -actin-Rq	CTCCTGCTTGCTGATCCACAT	
mcp-Fq	CCCGTGGGTTGGTTTACCAA	GU256635
mcp-Rq	CCGCCAGCAGTTTAATCTGG	
mx-Fq	TAAATGGCTGGGGTCGGGG	[9]
mx-Rq	CATTGCACGGAACGACCACC	
ifn- γ -Fq	AGATCAGAGGCTTTCAAATCCC	[9]
ifn- γ -Rq	CAACATGTGGCTAATCAGCTT	
irf-3-Fq	CCACTGCTCATCCCATGGCT	XM_038735464.1
irf-3-Rq	GAGGAGTCCTGTCTTAAACC	

decreased mortality of rainbow trout juvenile [6]. Arctiin exhibited dose-dependent inhibition of *Micropterus salmoides* rhabdovirus (MSRV) infection in grass carp cells and improved the survival rates of infected *Micropterus salmoides* [7]. Nervous necrosis virus (NNV) induced S-phase arrest in Grouper Fin-1 (GF-1) cells, and emodin treatment reduced intracellular ATP synthesis in infected GF-1 cells, thereby affecting viral proliferation [8]. Saikosaponin C significantly increased the largemouth bass survival rate in LMBRaV infected, maintained cellular shape, and showed inhibitory actions against LMBRaV growth in EPC cells [4]. Moreover, esculin mitigated LMBRaV-induced nuclear damage and enhanced the immune regulatory capacity of fish at the organismal level, exhibiting promising therapeutic and preventive properties [9].

Typically, these herbs or their extracted natural compounds are employed to feed in the form of additives in aquaculture [10,11]. A growing body of research on traditional Chinese medicinal herbs as aquaculture feed supplements indicates their multifaceted benefits. Beyond disease treatment, they enhance feed palatability, bolster fish immunity, protect liver and gallbladder functions, promote growth, and improve meat quality. As a result, this field has emerged as a new growth point in aquaculture pharmaceutical development [12–16].

Andrographolide (Andro), an active ingredient obtained from *Andrographis paniculata*, display an extensive range of pharmacological characters like anticancer, antiviral, anti-inflammatory, antibacterial, antitumor, antidiabetic, hepatoprotective, and immunomodulatory effects [17–19]. As a potent antiviral agent, Andro not only induces apoptosis in cancer cells but also inhibits delayed-type hypersensitivity, possesses antioxidant properties, and enhances the activity of cytotoxic T lymphocytes and natural killer cells [20]. In aquaculture, dietary Andro supplementation enhances antioxidant capacity, immune regulation, and intestinal flora balance of eels while promoting growth [21]. To explore the potential of Chinese herbal medicine in combating LMBRaV, this study screened fifteen naturally occurring compounds with defined chemical structures for antiviral activity against LMBRaV. Among these, Andrographolide demonstrated significantly potent antiviral effects. These findings offer substantial theoretical and practical implications for the development of antiviral drugs against LMBRaV and other economically important fish viruses in aquaculture.

2. Materials and methods

2.1. Cells, virus and fish

The EPC cells were cultured in Medium 199 (M199) with 10 % fetal bovine serum (FBS) at 25 °C. LMBRaV was preserved in our lab [22]. Largemouth bass (mean length: 8.525 $\pm$ 0.63 cm, mean weight: 7.673 $\pm$ 0.450 g) were collected from one breeding farm in Huangpi, Hubei Province, China. They were fed commercial feed thrice daily and housed in 200 L tanks maintained about 24 °C. To confirm the absence of

Table 2

Natural compounds utilized in this investigation, along with information on their Chemical Abstracts Service (CAS) number, and EPC concentration rate.

Compound name	CAS number	Safety Concentration
Astragaloside	17429-69-5	60 mg/L
Paeoniflorin	23180-57-6	60 mg/L
Fraxin	524-30-1	60 mg/L
Baicalin	21967-41-9	60 mg/L
trans-Cinnamic acid	140-10-3	40 mg/L
Liquiritigenin	578-86-9	40 mg/L
Forsythine	487-41-2	40 mg/L
Sinomenine	115-53-7	20 mg/L
Schizandrin B	61281-37-6	20 mg/L
Andrographolide	5508-58-7	10 mg/L
Forsythoside B	81525-13-5	10 mg/L
Saikosaponin A	20736-09-8	10 mg/L
Matrine	519-02-8	1 mg/L
Luteolin	491-70-3	1 mg/L
Dioscin	19057-60-4	0.5 mg/L

LMBRaV, five randomly selected largemouth bass were subjected to PCR testing using LMBRaV primers (Table 1). The experimental individuals were acclimated in another container at 24 °C for seven days prior to *in vivo* experiments. All animal experiments were approved by the Animal Experimental Ethical Inspection of Laboratory Animal, Yangtze River Fisheries Research Institute, Chinese Academy of Fishery Sciences (ID Number: YFI 2022-mengyan-201).

2.2. Cell toxicity of natural compounds

Fifteen natural chemicals (listed in Table 2) were purchased from Aladdin Reagent company (Shanghai, China). They were dissolved by Dimethyl sulfoxide (DMSO; Sigma, Germany), which the final concentration of DMSO was less than 5 %. Stock solutions were stored at –80 °C and subsequently diluted with M199 to achieve working concentrations. EPC cells were seeded in 96-well plate. After 24 h, 100 μ L serially diluted natural compounds (10, 20, 30, 40, 50, 60 mg/L) were added to each well, and the cells were grown for an additional 48 h. The plate was then incubated for one to 2 h after 10 μ L CCK-8 reagent (Beyotime, China) was added in each well according to the specification. Then the absorbance of above reaction was measured at 450 nm by a microplate reader to evaluate cytotoxicity of these natural chemicals. The value of cell viability was used to determine the vitality of cells and its calculation formula as following: cell viability = (OD₄₅₀ value of the drug-treated group/OD₄₅₀ value of the cell control group) $\times$ 100 %.

2.3. Andro inhibiting LMBRaV replication assay in vitro

Based on drug toxicity test results from Section 2.2, the maximum safe concentration for each natural compound was determined as the concentration resulting in cell viability $\geq$ 90 %. These concentrations were used for the initial screening of antiviral effects. The antiviral effect test was conducted as follows: EPC cells with the density of 1×10^5 cells/mL were cultivated in 12-well plate incubated at 25 °C for 12–24 h. Once formed a monolayer, the medium was removed, and the residual cells were washed by M199 medium for twice or three times. A diluted solution of LMBRaV (10^3 TCID₅₀) prepared in M199 was added into the cells, and they were incubated together for 2 h at 25 °C. Subsequently, the virus solution was removed. The residual cells were washed by M199 for 2 or 3 times and replenished fresh M199 medium the containing 5 % FBS and the natural compound to incubation continued at 25 °C for 48 h. The control group received a cell maintenance medium containing 0.5 % DMSO. Following incubation, the medium containing the natural compound was aspirated, and the residuum were washed by phosphate buffer saline (PBS) for 2 or 3 times. Then the cells were digested by 0.25 % trypsin, centrifuged for 5 min with 1000 rpm, and collected to extract total RNA. And finally, the *mcp* gene expression of LMBRaV were

investigated by RT-qPCR analysis.

2.4. Andro inhibiting LMBRaV replication assay in vitro

EPC cells were seeded in 12-well or 96-well cell culture plates until a monolayer were formed (about 12–24 h). LMBRaV, diluted in M199 ($10^3 \times \text{TCID}_{50}$), was added to above plates and incubated 2 h at 25 °C. After removing the mixture, each well was washed twice by PBS. Subsequently, Andro diluted by M199 containing 5 % FBS into different concentrations of 2, 3, 5, 8, 12.6, and 20 mg/L were added to the respective well to incubate for 48 h. Then the cell morphology was observed and documented by microscope. EPC cells in the 12-well plate were harvested and the LMBRaV *mcp* gene expression was quantified by RT-qPCR in order to analyze the Andro's impact on LMBRaV. The protective efficacy of Andro against LMBRaV infection was assessed based on the CCK-8 assay in 96-well plate according to the specification.

2.5. Viral binding assay

EPC cells (density of 1×10^5 cells/mL) were seeded into 25 cm² flasks to culture until reaching 75–90 % confluence. The LMBRaV (10^3 TCID_{50}) and Andro (20 mg/L) was added simultaneously into the cell culture flask to co-incubate at 4 °C for 60 min after the old culture medium removed. A group treated only with 5 mL of M199 medium containing LMBRaV (10^3 TCID_{50}) was set as control. The low-temperature incubation allowed for virus adsorption to the cells without initiating internalization [23]. Subsequently, the collected cell precipitates were washed by PBS for three times, and used to extract the total RNA followed by reverse transcription to cDNA. The cDNA as template was used to detect the content of the major coat protein gene (*mcp*) of LMBRaV by qPCR method.

2.6. Time-of-addition and time-of-removal assays

In time-of-addition experiment, LMBRaV ($10^3 \times \text{TCID}_{50}$) was added into cells in culture at a 12-well cell culture plate, and they were continued to cultivate together 2 h at 25 °C for to allow for viral attachment and entry. Subsequently, removed the medium containing virus, cells were washed twice by M199 before the fresh cell maintenance medium added. To evaluate the effect of Andro at different stages of viral replication, 20 mg/L Andro was added into above infected cells from 0 to 24 h every 2 h respectively. Andro (20 mg/L) was introduced for the time-of-removal experiments at 0 hpi. The medium drug-containing was discarded and then replenished fresh cell maintenance media at the same time points as the time-of-addition experiment. The LMBRaV *mcp* gene relative expression in all samples obtained at 24 h post-infection was detected by RT-qPCR. At every time point, three replications of each treatment were conducted.

2.7. Cell ultrastructure observation

EPC cells were seeded onto cell culture dishes in six-well plates. Upon reaching 70–85 % confluence, cells were infected with a dilution of LMBRaV (10^3 TCID_{50}) prepared in M199 for 2 h. Subsequently, a cell maintenance medium containing 20 mg/L Andro was added, and incubation continued at 25 °C for 48 h. Then cells treated with Andro were collected and fixed in 2.5 % glutaraldehyde overnight at 4 °C. After fixation, they were washed three times by PBS and dehydrated step by step through a gradient ethanol (30 %, 50 %, 70 %, 80 %, 90 % and 100 %). Dehydration was followed by stepwise de-alcoholization using 25 %, 50 %, and 75 % tert-butanol, with a final two exchanges in 100 % tert-butanol. Then the samples were dried in a dryer. Finally, the drying samples were fixed on stubs, sprayed by gold, and scanned by the Hitachi-SU8010 field emission scanning electron microscope (Hitachi, Japan).

Samples intended for internal structure observation were fixed in

osmium tetroxide, dehydrated, and embedded after being infiltrated in 2.5 % glutaraldehyde overnight at 4 °C. Next, ultrathin slices were cut by a Leica UC7 microtome (Leica, Germany) stained with lead citrate and uranyl acetate, and scanned by the Hitachi H7650 transmission electron microscopy (Hitachi, Japan).

2.8. Fluorescence observation of nucleus damage

For the nucleus damage assay, cells were seeded in confocal petri dishes and cultured for 12–24 h. LMBRaV (10^3 TCID_{50}) was introduced after a monolayer had formed, and it was left to infect for 2 h at 25 °C. After that, the viral solution was swapped out for a cell maintenance medium that contained 20 mg/L of Andro, and the incubation period was extended to 48 h. After three times washed by PBS, the EPC cells were fixed for 30 min with 4 % paraformaldehyde. Then they were stained by DAPI (1 mg/L) and Dil (5 mg/mL), and detected fluorescence using an Olympus FV3000 confocal microscope (Japan).

2.9. Assessing the anti-LMBRaV activity of Andro following different treatments

2.9.1. Pre-treatment with andro

EPC cells were diluted to a specific density and transferred to 12-well cell culture plate. After forming a monolayer, the old medium was discarded, and the residual cells were washed by M199 medium. Subsequently, they were supplemented a cell maintenance medium (M199 medium with 5 % FBS content) containing Andro (20 mg/L) and incubated at 25 °C for 6, 12, 24, 48, and 72 h respectively. The control group received a cell maintenance medium containing a specified volume of DMSO. Following incubation, the medium was removed, and LMBRaV (10^3 TCID_{50}) was added to continue culturing 2 h. The viral suspension was discarded, and fresh cell maintenance medium was added for 48-h incubation. After that, the cell samples were digested and collected, then the total RNA was extracted as above described. Finally, the total RNA was reverse transcribed into cDNA, and *mcp* gene expression of LMBRaV was detected.

2.9.2. LMBRaV and andro co-incubation assay

A LMBRaV (10^3 TCID_{50}) viral solution was combined with Andro (20 mg/L) in cell maintenance medium, mixed thoroughly, and incubated at room temperature for 0, 15, 30, and 60 min. Subsequently, the prepared LMBRaV-Andro mixture was added to 12-well plates containing confluent cell monolayers and incubated at 25 °C for 48 h. The total RNA was extracted and reverse transcribed into cDNA after cells harvested. The *mcp* gene expression was measured by RT-qPCR. Three separate runs of the experiment were conducted, with a virus-DMSO mixture acting as the control.

2.9.3. Andro post-processing

LMBRaV (10^3 TCID_{50}) virus suspension was added to 12-well plates containing confluent cell monolayers and cultured 2 h at 25 °C. Removed the virus suspension, the fresh cell maintenance medium was added into the plates. Cells were incubated for an additional 12 or 24 h. Subsequently, cells in plate of the medium discarded were washed twice by M199 medium. Cell maintenance medium with Andro (20 mg/L) was added into the cells, and continued to incubate for 24 h. Finally, total RNA of the cell samples was extracted, been reverse-transcribed into cDNA, as well as subjected to RT-qPCR analysis for LMBRaV *mcp* gene expression.

2.10. In vivo inhibition experiment

2.10.1. Infection modelling

LMBRaV was prepared in PBS at three concentrations: 10^2 , 10^3 , and 10^4 TCID_{50} . Each largemouth bass was administered 100 µL of virus dilution via intraperitoneal injection, while the control group received a

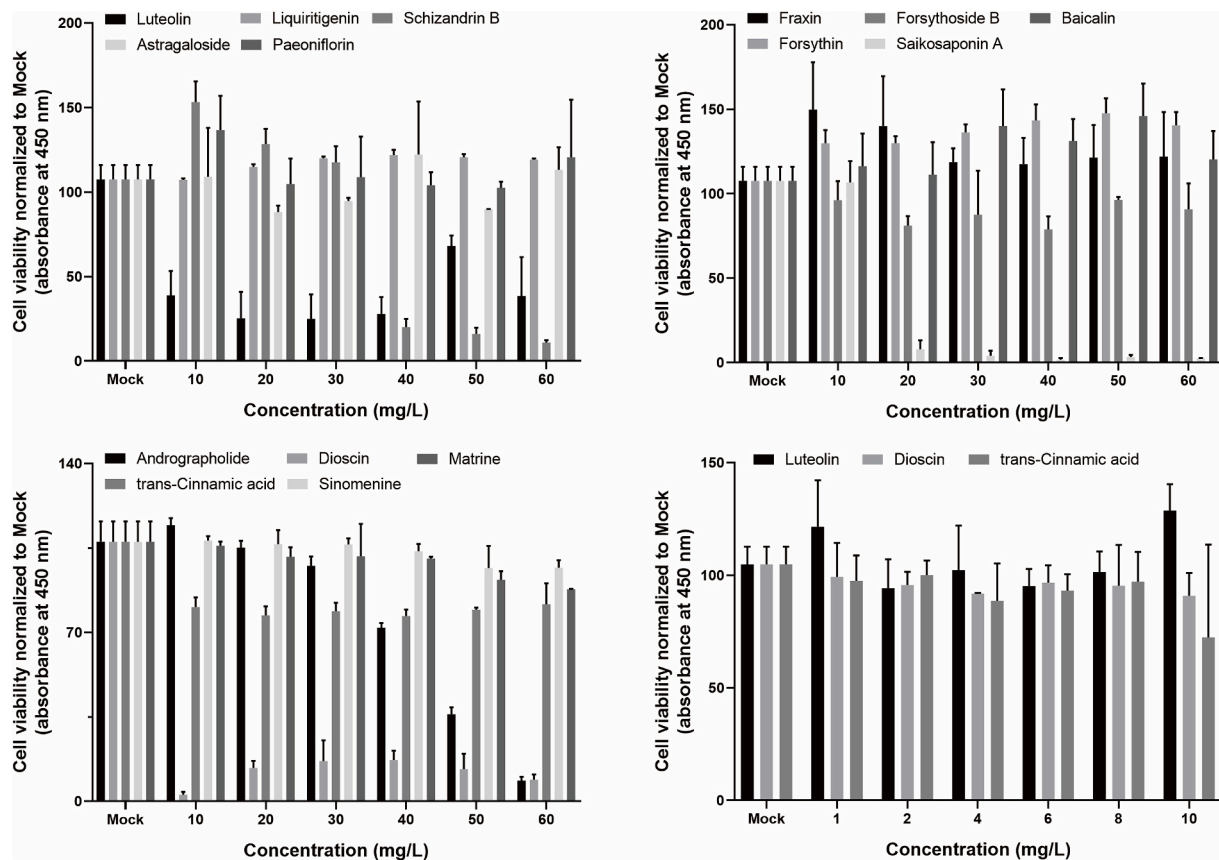


Fig. 1. Effects of various natural compounds at different concentrations on EPC cell viability. Values showed with normalized means $\pm$ standard deviations (SD) relative to the no-treatment group.

same volume of PBS. There were 20 individuals in each experimental group. They were raised in tanks with water temperature about 28 ± 0.5 °C. Death rates in each group were recorded for 14 days.

2.10.2. Andro toxicity test for largemouth bass

Prior to the experiment, commercial feed was supplemented with varying concentrations of Andro (0.1 %, 0.05 %, and 0.025 % M/M). The resulting feed mixtures were subsequently freeze-dried. Forty-eight largemouth bass were randomly divided into four groups (12 individuals per group): Control, Andro-0.1 %, Andro-0.05 %, and Andro-0.025 %. Each group was administered feed equivalent to 1 % of their body weight three times daily for seven days. Throughout the feeding period, fish health was monitored. Subsequently, three individuals of each group were randomly selected for anatomical observation to evaluate Andro's toxicity in largemouth bass. Based on these toxicity results, a concentration of 0.025 % Andro was selected for subsequent experiments.

2.10.3. Andro in vivo anti-LMBRaV activity

Largemouth bass were evenly divided into three groups (PBS, LMBRaV, and LMBRaV + Andro) of 25 largemouth bass per group with three replicates were set up. Fish in the LMBRaV and LMBRaV + Andro groups received an intraperitoneal injection of LMBRaV diluent (10^3 TCID₅₀, 100 μ L/fish), while the PBS group received an equivalent volume of PBS. Following the viral challenge, the LMBRaV + Andro group was fed commercial feed containing 0.025 % Andro, whereas the PBS and LMBRaV groups were fed regular commercial feed. All groups were fed an amount equivalent to 1 % of experimental fish body weight feed three times daily for 14 consecutive days. The experimental temperature was set at 28 ± 0.5 °C, and the oxygen concentration maintained at the same level as before the virus injection. Mortality rates were monitored

and recorded for each group over a fourteen-day period. Simultaneously, at 4, 7, and 14 days after viral injection, a random selection of fish was made for the same treatments. After removing the liver, spleen, and kidney tissues, total RNA extracted were reverse transcribed into cDNA. The virus dose and the expression levels of *mx*, *ifn- γ* , and *irf-3* in gene in every tissue were detected by RT-qPCR.

2.10.4. Effect of Andro on histopathology and gut microbiology of largemouth bass infected with LMBRaV

Following viral infection and Andro treatment as detailed in Section 2.9.3, three experimental fish were randomly selected from each group on day 7 post-treatment. Liver, spleen and head kidney were sampled for histopathological examination. Histopathological slide preparation was outsourced to Wuhan Pinuofei Biological Co., Ltd. Prepared slides were scanned and photographed by inverted microscope (Leica, Japan).

The midgut and hindgut of experimental fishes were carefully rinsed with sterile phosphate-buffered saline until no contents were discharged. Subsequently, bacteria adhering to the surface of the tissues were gently scraped off using a sterile microscope slide, collected in a 1.5 ml centrifuge tube, and immediately frozen in liquid nitrogen. Intestinal contents were collected as experimental samples, and total genomic DNA was extracted from the samples using a bacterial DNA extraction kit (Omega Biotek, USA). The extracted genomic DNA was analyzed using 1 % agarose gel electrophoresis. The purity and concentration of the DNA were measured using a Nanodrop ND 2000 spectrophotometer. The genomic DNA was used as a template to amplify the 16S rDNA V3-V4 region with the universal primers 338F 5'-(ACTCCTACGGGAGG-CAGCA)-3' and 806R 5'-(GGACTACHVGGGTWTCTA AT)-3'. Sequencing was performed on the Illumina MiSeq PE300 platform (Wuhan Tianyi Huayu Gene Technology Co.). The raw sequencing reads were subjected to quality control using Trimmomatic software. OTU clustering was

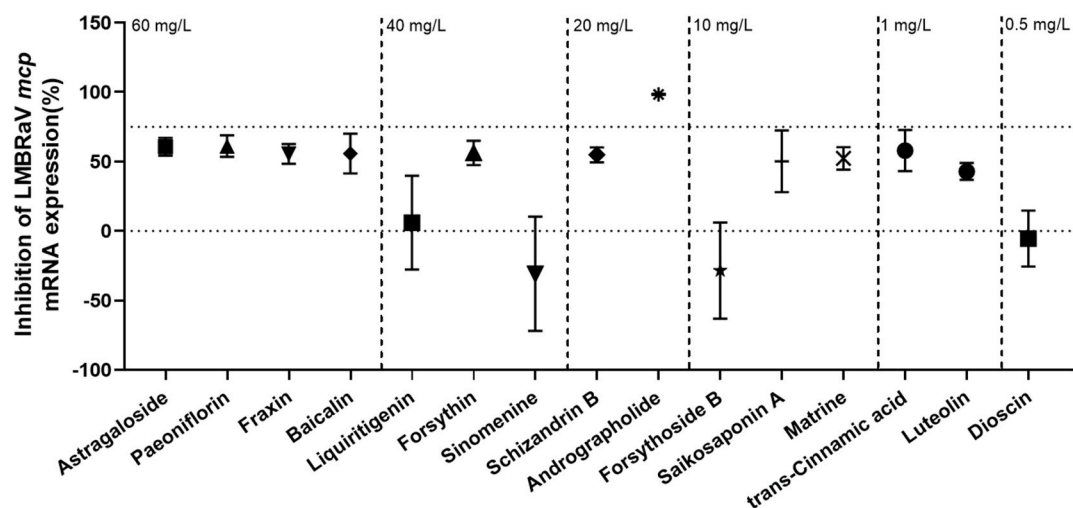


Fig. 2. Fifteen natural substances inhibit the expression of LMBRaV *mcp* mRNA. Data were shown as mean $\pm$ SD.

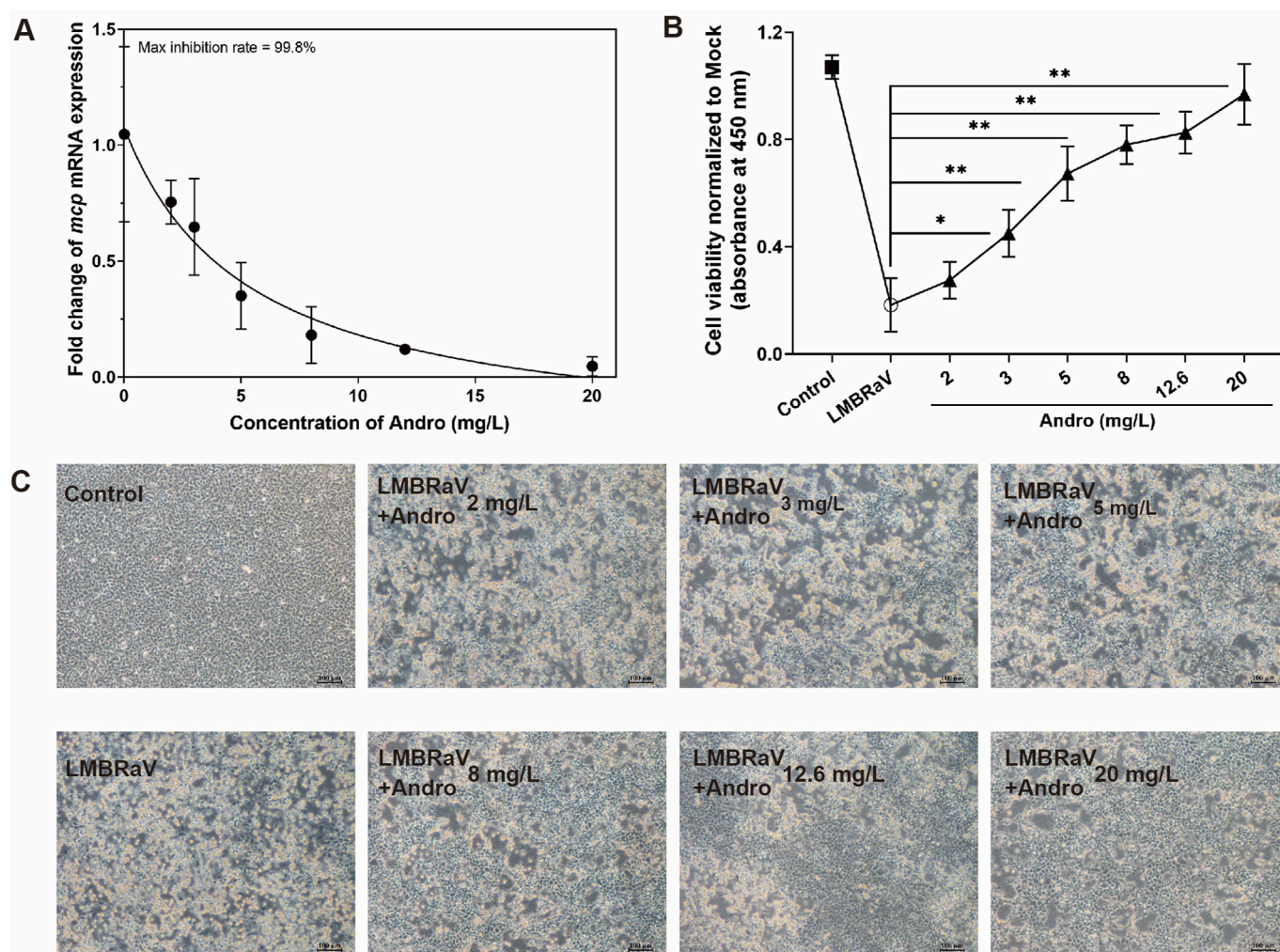


Fig. 3. Inhibitory effect of Andro on LMBRaV *in vitro*. (A) Six-point dose-response curve of Andro's anti-LMBRaV effect in EPC cell by *mcp* gene expression analysis. (B) Survival of LMBRaV-infected cells in EPC treated by Andro. (C) The cytopathic effect (CPE) of EPC cells caused by LMBRaV infection with different Andro concentrations. The error bars represent standard deviations. * p < 0.05, ** p < 0.01.

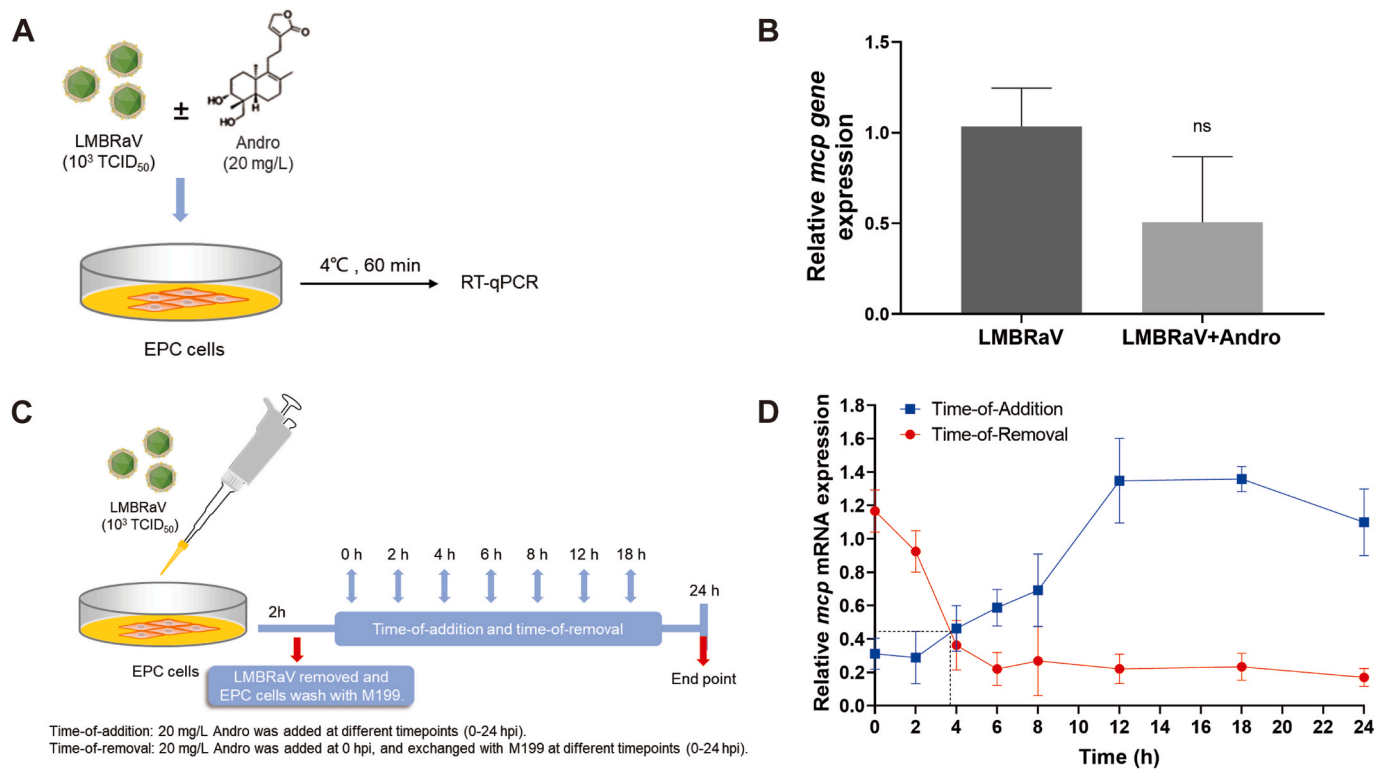


Fig. 4. Effects of Andro on the LMBRaV infection cycle. (A) Flow chart of the particle adsorption experiment. (B) Effect of Andro on the adsorption of LMBRaV with EPC cells. (C) Flow chart of the natural compound addition and removal experiment. (D) Effect of adding or removing Andro at different time points on the *mcp* expression of LMBRaV. Error bars indicated SD.

performed using UPARSE software (version 7.1, available at <http://drive5.com/uparse/>), with sequences clustered at a 97 % similarity threshold to generate an OTU table. Taxonomic classification of each sequence was conducted using RDP Classifier (<http://rdp.cme.msuedu/>).

2.11. Statistical analysis

The data were analyzed by statistical program GraphPad Prism 8. The findings were shown with mean value plus standard deviation (SD). When comparing two groups, independent sample t-tests were utilized, and when comparing several groups, one-way analysis of variance (ANOVA) was carried out. $p < 0.05$, statistical significance was established.

16S rRNA data analysis: α - and β -diversity analyses were carried out using QIIME software. α -diversity was assessed using indices such as Chao1 and Shannon, while principal component analysis (PCA) was employed to evaluate microbial diversity within the samples. Non-parametric tests were used to verify that Chao 1 and Shannon's data conformed to a normal distribution. When $P > 0.05$, the data were considered to conform to a normal distribution. Data that conformed to the normal distribution made the t-test and data that did not conform used the non-parametric Mann-Whitney test.

3. Result

3.1. Cell toxicity of natural compounds

A CCK-8 kit was employed to evaluate the cytotoxicity of natural compounds on EPC cells. The maximum safe concentration for each compound was determined when it made the cell viability exceed 90 %. These values are presented in Fig. 1 and Table 2. The fifteen natural compounds were categorized into six groups based on concentration: 60, 40, 20, 10, 1, and 0.5 mg/L for antiviral activity screening.

3.2. Antiviral effects of 15 natural compounds in vitro

Anti-LMBRaV natural compound screening was performed using the maximum safe concentration of each compound, as determined by the EPC toxicity assay (Section 3.1). Screening results are presented in Fig. 2. Natural compounds that reduced LMBRaV *mcp* gene expression in EPC cells by ≥ 75 % were considered to exhibit a significant anti-LMBRaV effect. Among the tested compounds, Andro at concentration of 20 mg/L demonstrated the highest inhibition ratio of 98.38 % against LMBRaV *mcp* gene expression, prompting its selection for further experimental investigation.

3.3. Antiviral effect of Andro in vitro

To further investigate Andro's antiviral effects, the *mcp* gene expression of LMBRaV at six concentrations of Andro's was detected by RT-qPCR. As illustrated in Fig. 3A, Andro exhibited a concentration-dependent antiviral effect *in vitro*, with the lowest *mcp* expression (0.2 %) observed at a concentration of 20 mg/L. Moreover, Andro demonstrated a protective effect on EPC cell viability, increasing the growth viability of LMBRaV-infected EPC cells by 78.62 % at 48 h post-infection (Fig. 3B). The cytopathic effect (CPE) caused by an LMBRaV infection was dramatically decreased by 20 mg/L of Andro, as demonstrated by microscopic inspection (Fig. 3C).

3.4. Effects of Andro on LMBRaV infection cycle

The experimental procedures are depicted in Fig. 4A and C. Results from the particle adsorption assay revealed no significant difference in *mcp* expression between Andro-treated infected cells and the LMBRaV-only group (Fig. 4B). This indicated that Andro did not significantly affect LMBRaV particle adsorption onto EPC cells. To elucidate Andro's impact on the LMBRaV infection cycle, we examined viral proliferation

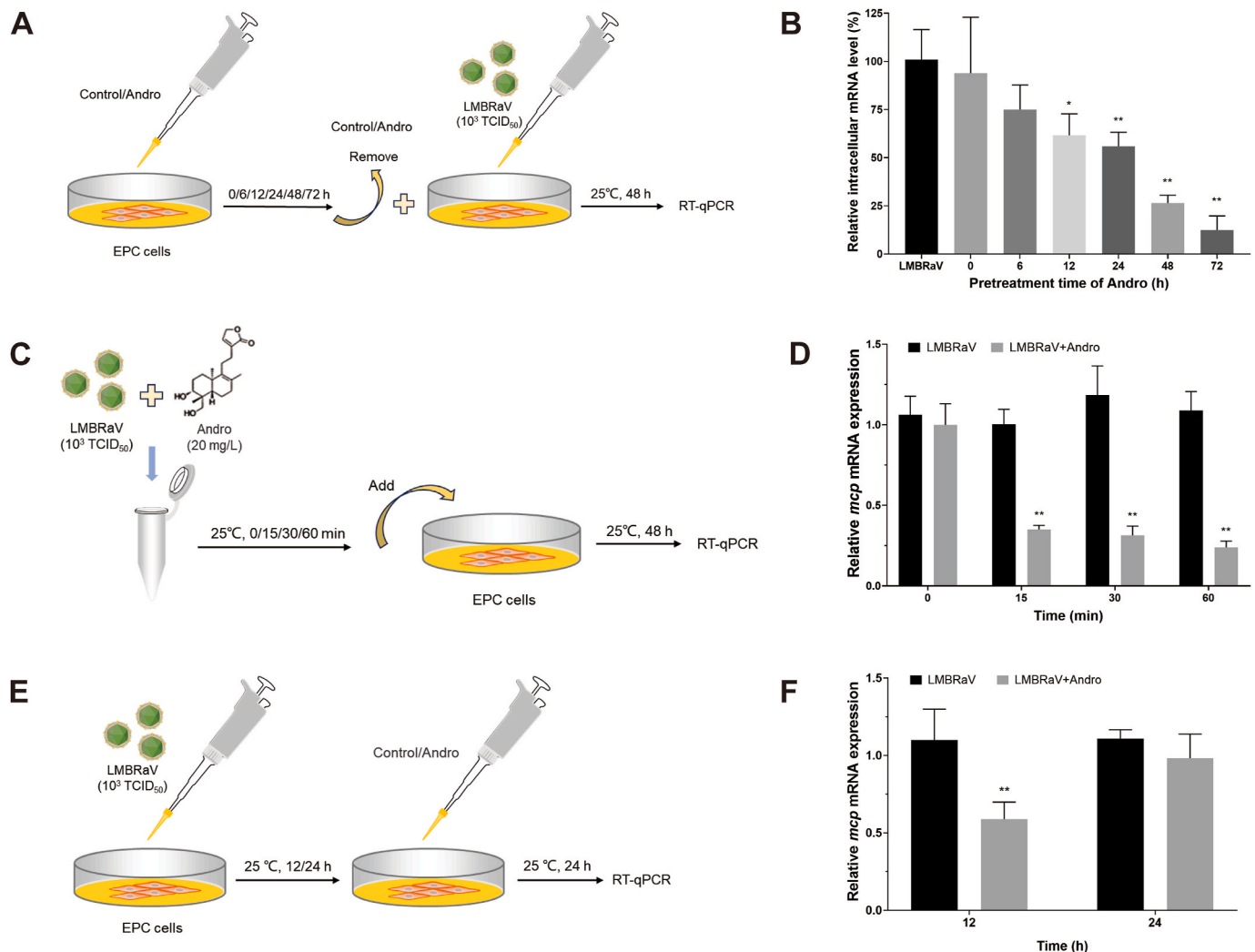


Fig. 5. Effect of different Andro's treatments on LMBRaV proliferation. (A) Flow chart of the pre-treatment process. (B) Detection results of the *mcp* expression in EPC cells infected with LMBRaV after different durations of Andro pre-treatment. (C) Flow chart of drug-virus co-incubation experiment. (D) Detection results of the *mcp* expression in EPC cells in drug-virus co-incubation assay. (E) Flow chart of the post-treatment process. (F) Effect of post-treatment on the *mcp* expression in EPC. Normalized to no-treatment values, each value presented as mean $\pm$ SD. ** $p < 0.01$, * $p < 0.05$.

following addition or removal drug at various time points. As shown in Fig. 4D, the *mcp* expression gradually increased with delayed Andro addition, culminating in a sharp 65.67 % increase at 8–12 h. Similarly, the *mcp* expression increased with delayed Andro removal. Conversely, the *mcp* expression progressively decreased as Andro removal was postponed, with a pronounced reduction observed at 2–4 h. Based on the overlapping timeframes of these experiments, Andro appeared to interfere with viral invasion during the 2–4 h period.

3.5. Effects of different treatments of Andro on LMBRaV proliferation

Experimental procedures are depicted in Fig. 5A, C, and 5E. Fig. 5B illustrated the impact of Andro pre-treatment on LMBRaV proliferation in EPC cell. It indicated that preincubation EPC cells with Andro for 12, 24, 48, and 72 h prior to LMBRaV infection significantly reduced viral proliferation. As shown in Fig. 5D, the *mcp* expression in LMBRaV-infected EPC cells co-incubated with Andro for 15, 30, and 60 min decreased significantly, indicating a direct antiviral effect of Andro. The most pronounced virucidal activity was observed after 60 min of co-incubation, with a 96.26 % reduction the *mcp* gene expression in intracellular. Fig. 5F presented the results of post-treatment with Andro. The data indicated that Andro retained significant antiviral activity up

to 12 h post-LMBRaV infection, with a 51.14 % reduction of the *mcp* gene expression compared to control group. However, Andro exhibited no antiviral effect when administered 24 h post-infection, suggesting that its therapeutic efficacy is limited to the early stages of infection.

3.6. Cell ultrastructure observation

Using scanning electron microscopy, the EPC cell surface morphology with LMBRaV-infected and the drug's protective effect were observed and depicted in Fig. 6A. EPC cells in control group exhibited a smooth surface and intact morphology. In contrast, cells infected with LMBRaV for 48 h displayed cellular damage characterized by surface disruption and pore formation. Treatment with Andro preserved the overall cellular morphology of infected cells, with no significant deviations from control group. According to these findings, it was suggested that Andro effectively prevented LMBRaV-induced cellular damage.

Transmission electron microscopy observations revealed the destructive effect of LMBRaV infection on the surface structure of EPC cell, and the protective role of Andro in LMBRaV infected cell. As depicted in Fig. 6B, LMBRaV-infected EPC cells exhibited significant morphological alterations, including nuclear wrinkling and

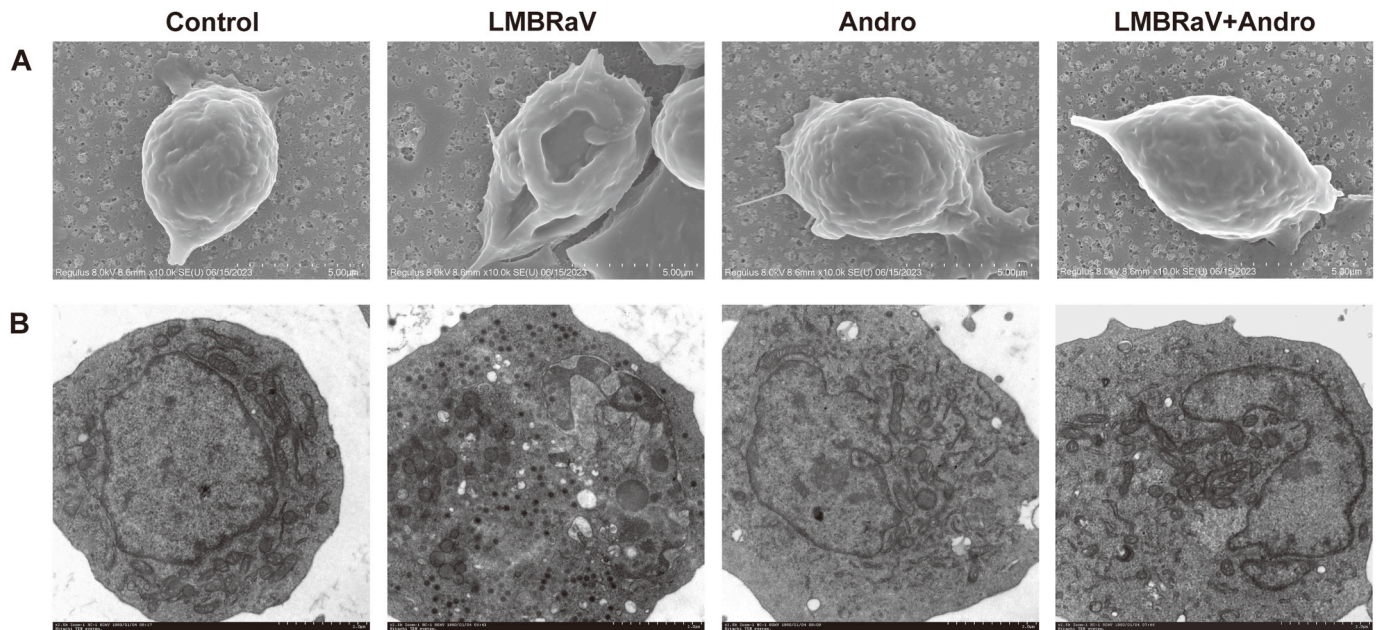


Fig. 6. Observations of the surface and internal ultrastructure of EPC cells. (A) Andro protects the surface structural morphology of EPC cells from LMBRaV. (B) Andro protects the internal structure and morphology of EPC cells from LMBRaV.

deformation, cytoplasmic vacuolation, and mitochondrial damage. These observations suggest that LMBRaV infection compromised the normal structure and function of EPC cells, potentially leading to abnormal cell behavior or death. In contrast, EPC cells treated with Andro maintained the morphology comparable to the control group, with intact nuclei, cytoplasm, and mitochondria. These findings indicate that Andro effectively protected EPC cells from the structural and functional disruptions induced by LMBRaV infection.

3.7. Cell nuclear damage observation

Cells were stained with a DAPI/Dil staining solution to assess nuclear damage induced by LMBRaV and the impact of drug treatment. Results are presented in Fig. 7. After 48 h of LMBRaV infection, EPC cells exhibited a rounded morphology with swollen and fragmented nuclei. In contrast, cells treated with Andro maintained the nuclear and cell membrane morphology similar to the control group, exhibiting normal staining patterns. These findings indicate that Andro effectively protected against LMBRaV-induced nuclear damage and demonstrated potent anti-LMBRaV activity.

3.8. Andro anti-LMBRaV activity assay in vivo

3.8.1. LMBRaV challenge assay

Survival rates of largemouth bass infected with various LMBRaV titers are presented in Fig. 8A. Fish infected with a viral titer of 10^4 TCID₅₀ exhibited initial mortality on day 2, culminating in a 100 % cumulative mortality rate by day 14. For fish infected with 10^1 and 10^2 TCID₅₀ of LMBRaV, mortality onset occurred on day 4, while those infected with 10^3 TCID₅₀ LMBRaV experienced mortality onset on day 3, resulting in final cumulative mortality rates of 21 %, 50 %, and 77 %, respectively. A linear regression analysis (Fig. 8B) revealed a strong positive correlation ($y = 26.4x - 4$, $R^2 = 0.9974$) between largemouth bass mortality and LMBRaV dose. Considering the reliability and reproducibility of these results, a viral titer of 10^3 TCID₅₀ was selected as the challenge dose for subsequent experiments.

3.8.2. Safe concentrations of andro on largemouth bass

Among largemouth bass fed diets containing varying concentrations

of Andro, those in 0.025 % Andro-group showed decreased food intake while those in the other groups displayed no abnormal symptoms. To further assess the potential toxicity of Andro in largemouth bass, three individuals randomly selected from each group were dissected and examined visceral organs. As depicted in Fig. 9, largemouth bass in Andro-0.1 % and Andro-0.05 % groups displayed hepatic abnormalities such as liver whitening compared to the control group. There were no obvious differences observed in Andro-0.025 % group. The results suggested that higher Andro concentrations may adversely affect largemouth bass health. Consequently, a feeding concentration of 0.025 % Andro was selected for subsequent experiments.

3.8.3. Andro anti-LMBRaV activity in vivo

Fig. 10 illustrated the therapeutic efficacy of Andro in treating largemouth bass with LMBRaV infection. In control group, the cumulative mortality rate reached 65 % when infected with LMBRaV. Compared with this, the Andro-treated group exhibited a 40 % reduction in mortality, with the peak mortality occurring primarily on the 10th day post-infection. According to RT-qPCR analysis (Fig. 10C), liver, spleen, and kidney *mcp* gene expression was significantly reduced in Andro-treated largemouth bass at the 4, 7, and 14 day post-infection periods.

3.8.4. Effect of Andro on antiviral-related genes expression in LMBRaV-infected largemouth bass

The antiviral-related gene expression of *mx*, *ifn-γ*, and *irf-3* in LMBRaV-infected largemouth bass following Andro treatment were detected and depicted in Fig. 11. Compared to the LMBRaV-only group, Andro feeding for 4, 7, and 14 days induced varying degrees of upregulation of these antiviral genes in spleen, kidney and liver of infected fish. The most pronounced upregulation of immune genes occurred at 4 and 7 day post-treatment, suggesting that Andro primarily exerted its antiviral effects by influencing LMBRaV replication during the early infection stage.

3.8.5. Andro on LMBRaV-induced histopathological damage in largemouth bass

As illustrated in Fig. 12, Andro mitigated histopathological changes in LMBRaV-infected largemouth bass. The liver of fish in the LMBRaV-

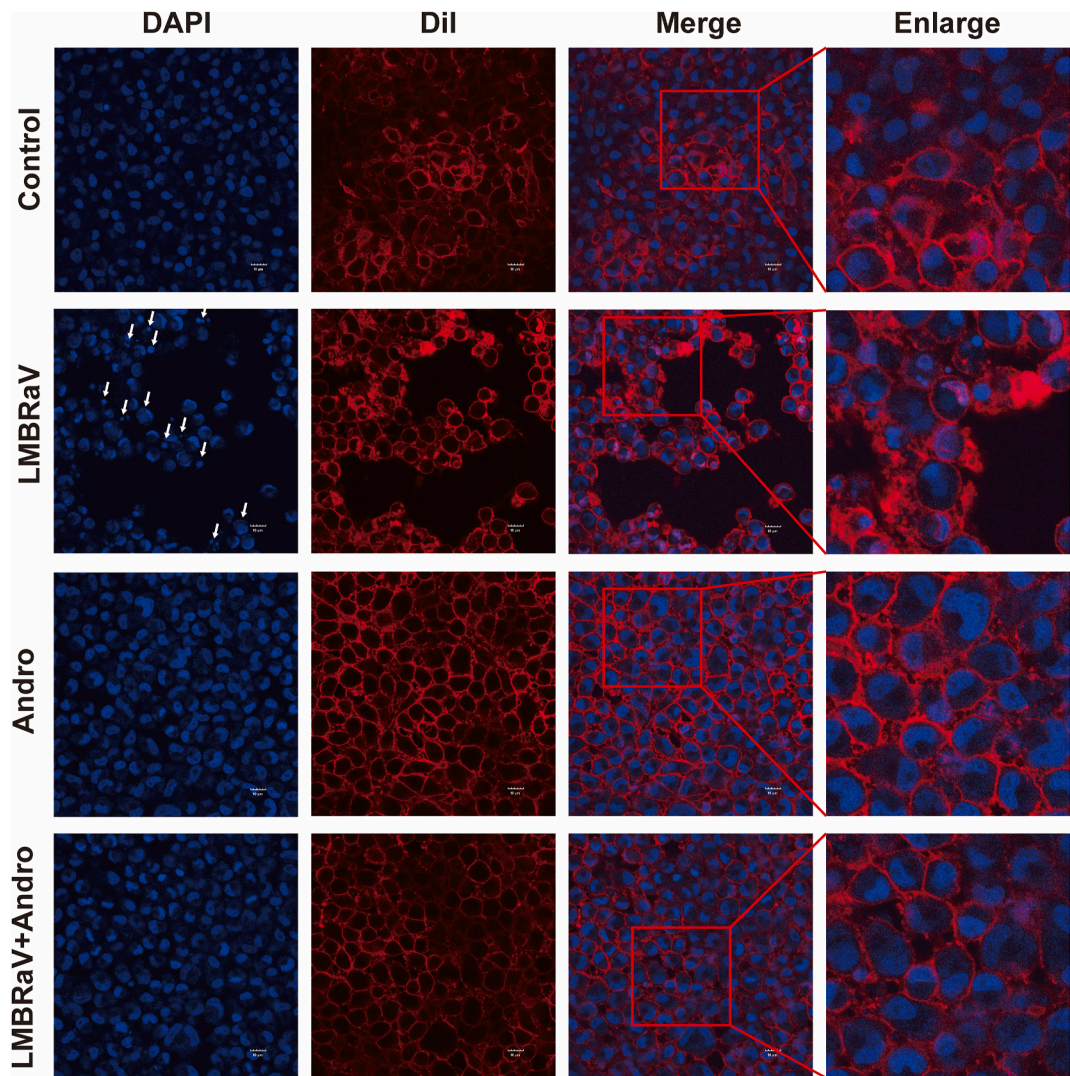


Fig. 7. EPC cells are protected by Andro against LMBRaV-induced nuclear damage.

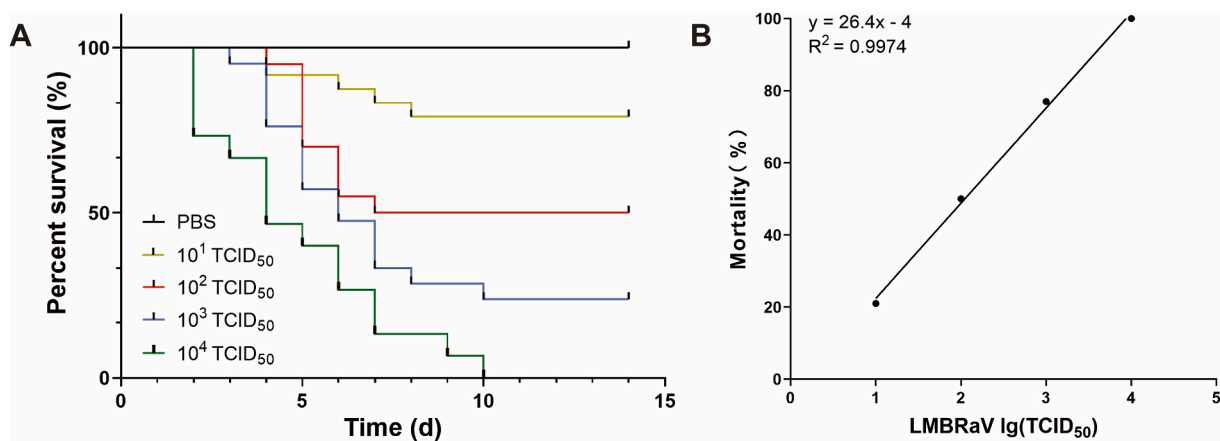


Fig. 8. Pathogenicity of LMBRaV against largemouth bass. (A) Largemouth bass survival curves after intraperitoneal injection LMBRaV of 10^1 , 10^2 , 10^3 and 10^4 TCID₅₀/mL respectively. (B) Linear correlation between mortality of largemouth bass and viral dose. n = 20.

infected group exhibited severe necrosis, characterized by hemorrhagic spots and inflammatory exudate. Splenic tissue displayed dispersed melanin macrophages, indistinct boundaries between red and white pulp, and pronounced vacuolation within the white pulp. Additionally, inflammatory infiltration was observed in kidney tissue. As opposed to

the control group, tissues of LMBRaV-infected largemouth bass did not show significant differences after treatment with Andro. These results suggested that Andro effectively reduced virus-induced pathological damage to the visceral organs of largemouth bass.

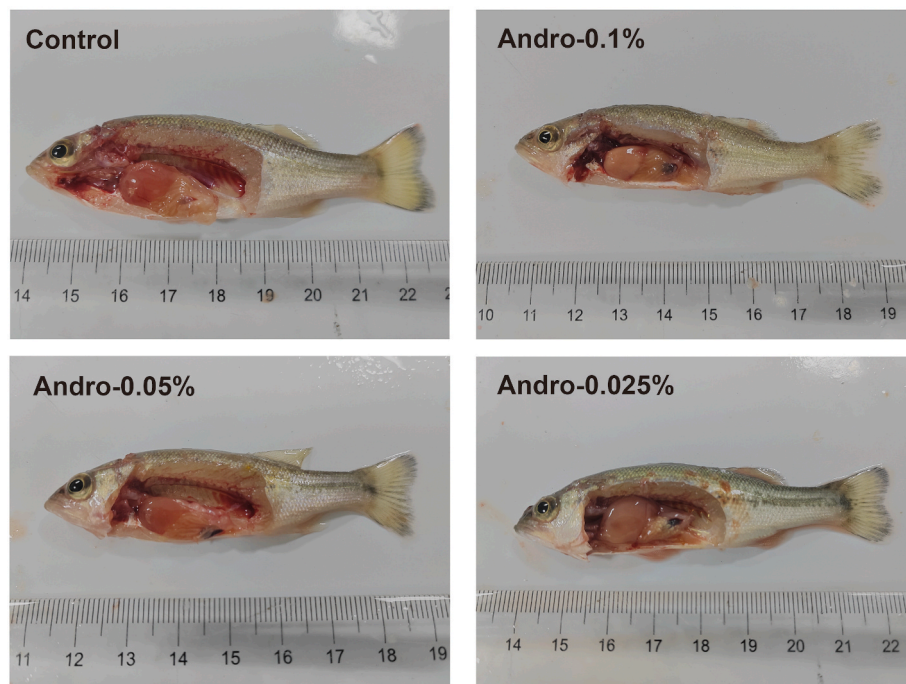


Fig. 9. High concentrations of Andro caused internal visceral damage in largemouth bass.

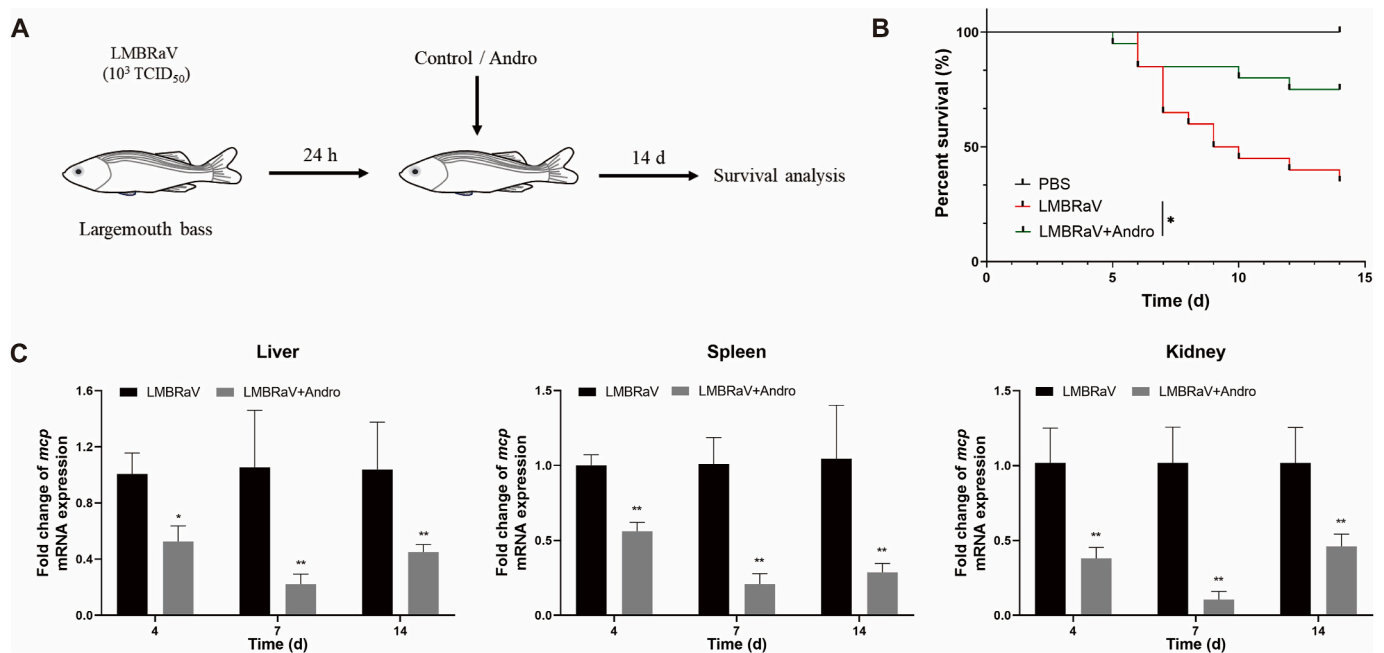


Fig. 10. Inhibitory effect of Andro on LMBRaV *in vivo*. (A) Flow chart of the pre-treatment process. (B) Significantly increased survival rate of LMBRaV-infected largemouth bass treated by Andro. (C) Significantly reduced *mcp* gene relative expression in the liver, spleen and kidney of LMBRaV-infected largemouth bass. Error bars indicated the significance between different groups. $n = 25$. * $p < 0.05$, ** $p < 0.01$.

3.8.6. Andro on gut microorganisms of largemouth bass infected with LMBRaV

An analysis of 16S rRNA genes of gut microorganisms with LMBRaV-infected largemouth bass was conducted to determine whether Andro impacted the community of microbes in the gut. Results were presented in Fig. 13. Chao1 and Shannon indices revealed a significant increase following post-infection Andro supplementation (Fig. 13A and B), indicating enhanced microbial richness and diversity. Principal Coordinate Analysis (PCoA) visualized microbial relationships within and

between groups (Fig. 13C), demonstrating distinct microbial distributions among the three intestinal groups and highlighting the effects of LMBRaV and Andro treatments. At the phylum level, *Fusobacteriota* dominated (69.83 %) in healthy largemouth bass, followed by *Firmicutes* (11.45 %), *Proteobacteria* (5.99 %), and *Actinobacteriota* (4.5 %) (Fig. 13E). LMBRaV infection decreased the relative abundance of major beneficial bacteria, such as *Fusobacteriota* and *Actinobacteriota*. However, an increase of the dominant beneficial bacteria, such as *Actinobacteriota* and *Verrucomicrobiota* was observed in LMBRaV-infected

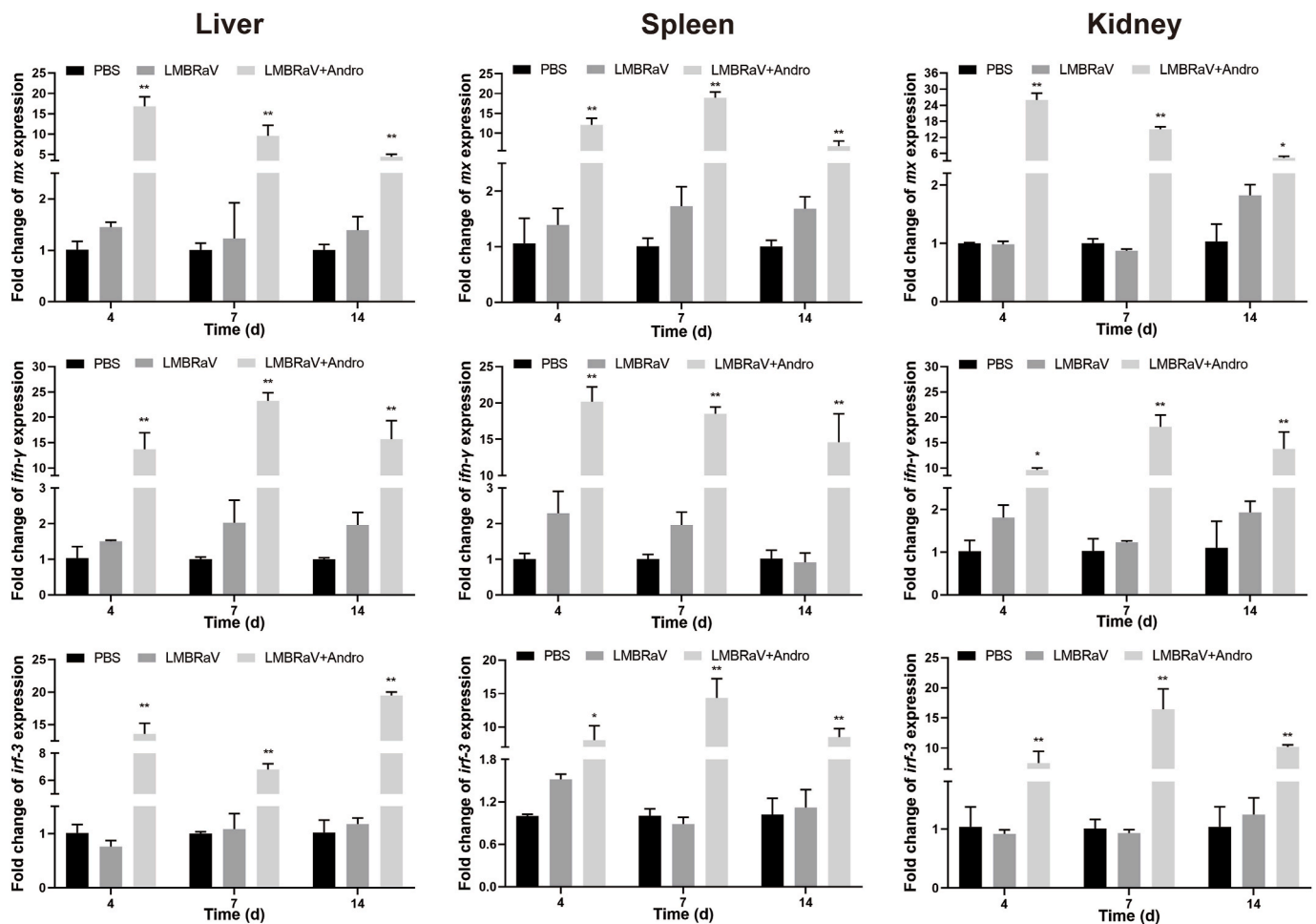


Fig. 11. Andro upregulated the antiviral-related genes expression of *mx*, *ifn-γ*, and *irf-3*. Each value presented as a mean $\pm$ SD, normalized to the no-treatment group values. * $p < 0.05$, ** $p < 0.01$.

individuals after the Andro treatment. At the bacterial genus level (Fig. 13F), *Mesomycolasma* significantly increased in the LMBRaV infection group but decreased in the Andro-treated group. Additionally, beneficial bacteria like *Akkermansia* increased in relative abundance following Andro treatment in LMBRaV-infected specimens.

4. Discussion

The development, synthesis, and application of pharmacologically active monomers extracted from natural plants have great prospects in aquaculture. Utilizing cell models for antiviral drug screening offers advantages including low cost, controlled experimental conditions, and high reproducibility, making them essential tools in viral disease research. The complex aquaculture environment facilitates the rapid spread of LMBRaV, severely hindering the sustainable development of largemouth bass aquaculture. Consequently, the discovery of novel antiviral medications to control LMBRV infections is imperative. Through screening fifteen natural compounds against LMBRV, Andro was identified as a promising candidate exhibiting potent antiviral activity. Further investigation of Andro's anti-LMBRV effects at varying concentrations revealed consistent efficacy and a concentration-dependent antiviral response. Notably, 20 mg/L Andro demonstrated a maximum inhibition rate of 99.8 % against LMBRV.

Viruses are primarily inactivated by three mechanisms: direct inactivation, interference with its attachment to cells, and antiviral inhibition its replication and biosynthesis of host cell. [24]. Previous research divided the LMBRaV infection process in EPCs into five stages: viral

attachment (0–2 h), viral invasion (2–4 h), cytoplasmic transport (4–12 h), viral replication (12–24 h), and viral assembly and release (post-24 h) [25]. Particle adsorption experiments indicated that Andro did not prevent viral attachment to host cells. However, introducing Andro to cells at 2–4 h or 4–6 h post-LMBRaV infection significantly reduced the *mcp* gene expression in EPC cell, suggesting antiviral activity during the viral invasion stage. Additionally, co-incubation experiments demonstrated Andro's direct virucidal effects. This study found that Andro effectively controlled LMBRaV proliferation, providing some preventive effect against LMBRaV infection. Moreover, Andro exhibited therapeutic potential in treating LMBRaV-infected cells, especially in cases of less severe cellular pathology. Andro significantly protected EPC cell morphology and internal structure from LMBRaV infection, preventing virus-induced cellular damage and death. This preservation maintained normal cell growth, surface integrity, and internal structure.

The present study also revealed that the inoculation dosage of LMBRaV exhibited a positive correlation with the final mortality rate of largemouth bass during the infection experiment. Subsequently, a drug treatment experiment involving Andro revealed a significant therapeutic effect of Andro against LMBRaV infection in largemouth bass ($p < 0.05$). Through the virus infection model, it was found that the mortality rate of largemouth bass sharply increased on day 4 post-LMBRaV infection, indicating a rapid proliferation phase of the virus during this period. In the antiviral process, drugs typically achieve their therapeutic effects by activating or regulating the antiviral response of host [26]. Given that LMBRaV primarily proliferating in liver, spleen, and kidney of largemouth bass [27], samples from these three key organs were

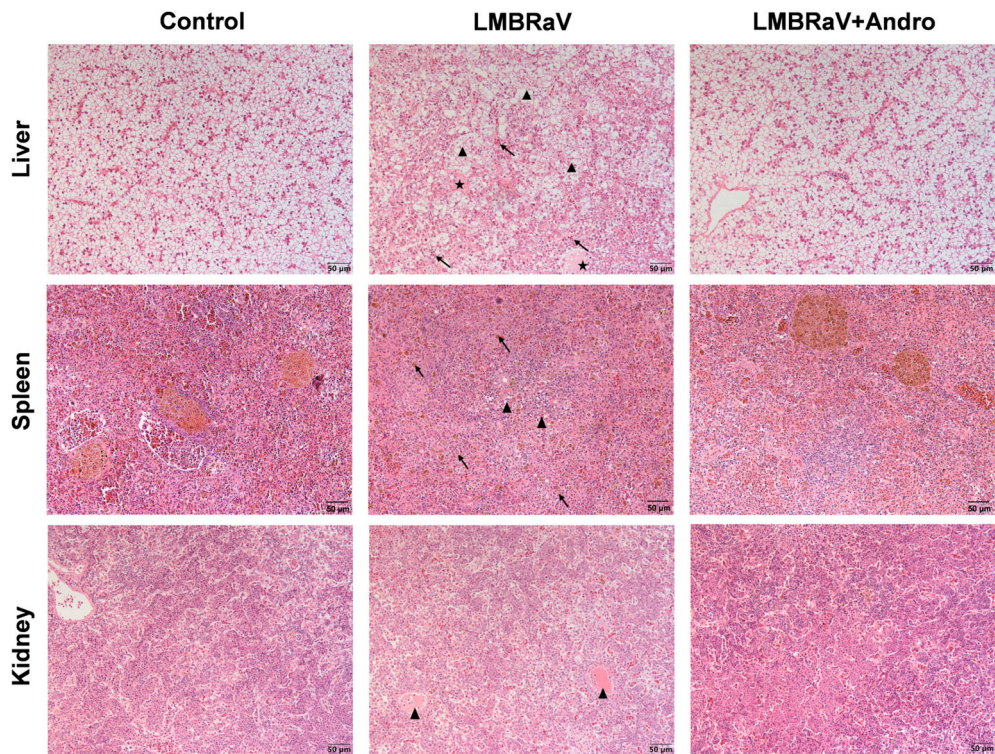


Fig. 12. The effect of Andro on histopathological changes induced by LMBRaV infection in largemouth bass tissues. For the examination of liver tissues from largemouth bass, arrows denote hemorrhage, triangles indicate liver tissue necrosis, and asterisks represent inflammatory exudation. Regarding spleen tissue pathology, arrows denote dispersed melanomacrophages, while triangles signify vacuolization. In renal tissue pathology, arrows depict inflammatory exudation, with triangles indicating thinning and detachment of blood vessel walls.

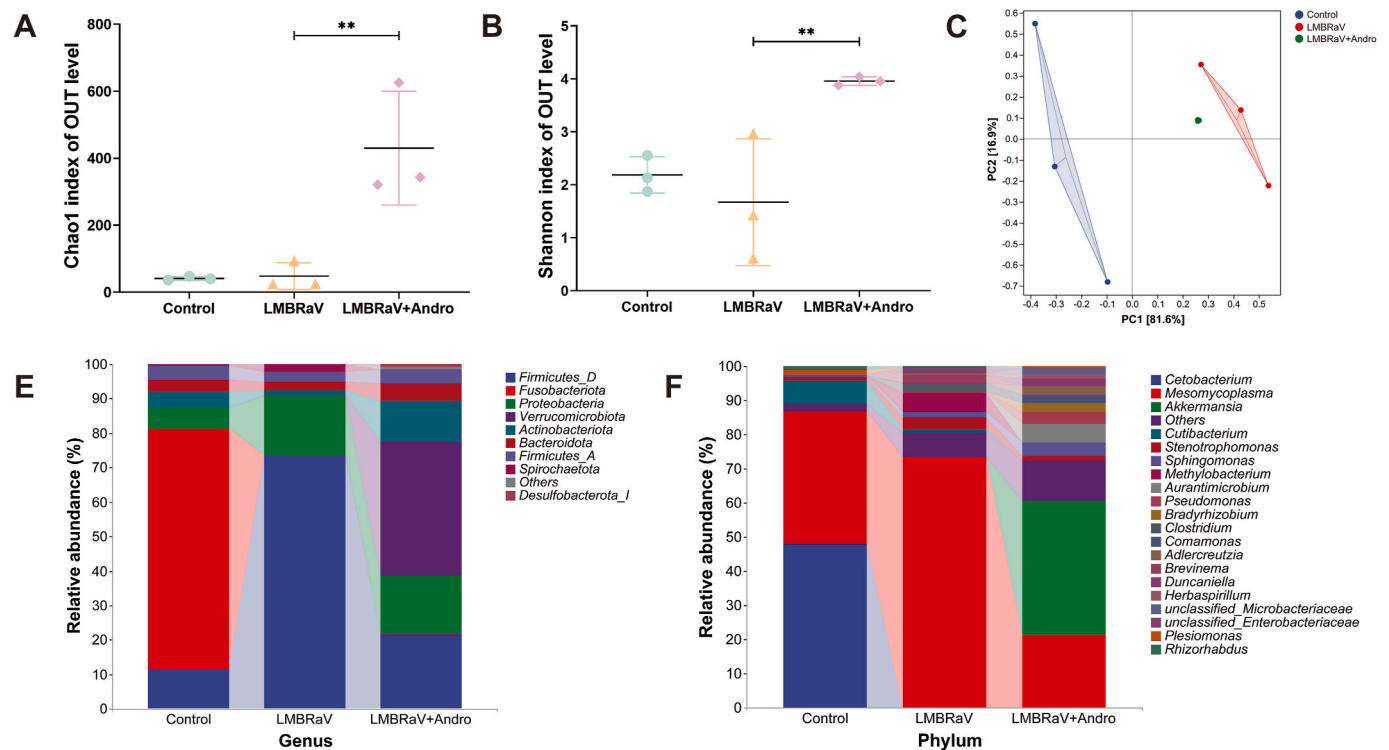


Fig. 13. Changes of intestinal microbiota in largemouth bass after Andro treatment. (A) Chao1 index. (B) Shannon index. (C) PCoA analysis. (D) Phylum-level changes in intestinal microbiota composition. (E) Genus-level changes in intestinal microbiota composition. (F) Changes of Phylum-level composition in intestinal microbiota. Chao1 and Shannon exponential analyses were carried out using a non-parametric test. $**p < 0.01$.

collected during the virus proliferation phase to analyze the antiviral-related immune genes expression. This provided direct evidence of virus-host interactions. The Mx protein, as the expression product of interferon-inducible genes, has become a focus of research in the field of innate immune effector molecules [28]. When interferon was induced, Mx protein was also activated, exhibiting its antiviral effect. By monitoring the effect of drugs on *mx* mRNA expression, we could indirectly evaluate their regulatory effect on the host's antiviral capacity. As part of the interferon regulatory transcription factor family, IRF-3 plays an important role during viral infection. It can activate the transcription of a series of antiviral genes, thereby enhancing the host's defense mechanism against viruses [29]. Therefore, a drug capable of regulating the antiviral response by influencing IRF-3 activity could serve as a promising antiviral therapeutic. IFN- γ , as the sole type II interferon plays a crucial role in coordinating the innate and adaptive immune responses. IFN- γ not only directly inhibits viral replication but also enhances overall antiviral capacity by modulating the function of other immune cells [30]. Consequently, monitoring the effect of drugs on expression levels can provide valuable insights into the mechanism of drug antiviral action. Experimental findings from this research demonstrated that Andro significantly induced rapid expression of antiviral-related gene of *mx*, *ifn- γ* , and *irf-3*, activating the innate immune response of largemouth bass. This process contributed to controlling LMBRaV proliferation in the liver, spleen, and kidneys, thereby reducing viral replication and infectivity during the infection process. During toxicity testing, pathological changes occur in the organs of experimental animals due to toxic effects. Histopathological examination of tissue sections can reveal pathological alterations within each organ at the microscopic level. Observations of liver, spleen, and kidney tissue sections from LMBRaV-infected largemouth bass indicated that Andro could prevent pathological damage induced by LMBRaV infection.

Microorganisms play indispensable roles in host growth, development, nutritional metabolism, and immune function [31]. Gut microbiota and their hosts have symbiotic and parasitic relationships [32]. In most cases, gut microbiota inhibit viruses by destabilizing viral particle morphology and preventing viral internalization and adsorption [33]. However, gut microbiota can also interact directly with pathogens, facilitating viral invasion or replication [34]. For instance, bacterial surface polysaccharides can interact with poliovirus, enhancing virus particle stability and binding to host cell surfaces [35,36]. Additionally, symbiotic bacteria could inhibit antiviral immune responses, including type III interferons, thereby facilitating murine norovirus infection [37]. Previous research identified *Fusobacteriota*, *Proteobacteria*, *Firmicutes*, and *Actinobacteriota* as the dominant bacterial phyla in intestinal tract [38,39], which was consistent with our findings. The Gram-positive bacteria *Actinobacteriota* can reportedly produce biologically active secondary metabolites to reduce pathogenic virulence [40]. Within *Verrucomicrobiota*, certain members exhibit probiotic characteristics by enhancing adaptive immune response and reducing inflammation [41]. In this study, the dominant bacterial phyla in the intestinal tract of largemouth bass included *Fusobacteriota*, *Firmicutes*, *Proteobacteria*, and *Actinobacteriota*. *Actinobacteriota* abundance decreased following LMBRaV infection but increased in the LMBRaV + Andro group. Similarly, *Verrucomicrobiota* abundance significantly increased in the LMBRaV + Andro group. Additionally, the relative abundance of *Spirochaetes*, common pathogens in humans and animals, induced intestinal inflammation [38,42], notably decreased in LMBRaV-infected largemouth bass treated with Andro. These findings suggest that Andro may increase beneficial bacterial abundance, decrease harmful bacterial abundance, and enhance fish immunity against pathogenic invasion.

5. Conclusions

This study demonstrated that Andro exhibited superior efficacy compared to 15 other natural compounds in inhibiting LMBRaV replication both *in vitro* and *in vivo*. Andro maintained normal cellular

morphology, reduced nuclear damage of LMBRaV-induced, and modulated LMBRaV replication by enhancing the expression of antiviral-related genes. Additionally, Andro increased the abundance of beneficial gut flora. These findings indicate that Andro possesses obvious anti-LMBRaV effect holding a potential drug development in aquaculture.

CRedit authorship contribution statement

Jingya Li: Contributed to conceptualization, methodology, data collection, formal analysis and writing-original draft. **Mingyang Xue:** Contributed the conceptualization, data collection and writing-original draft. **Chen Xu:** Responsible for overseeing the experimental design. **Nan Jiang:** Conducted the software analysis and methodology. **Yiqun Li:** Conducted the software analysis and methodology. **Yuding Fan:** Contributed to project administration and supervision. **Yong Zhou:** Contributed the funding acquisition, resources and supervision. **Yan Meng:** Contributed to funding acquisition, acquisition and writing – review & editing.

Ethics statement

All animal experiments were approved by the Animal Experimental Ethical Inspection of Laboratory Animal Center, Yangtze River Fisheries Research Institute, Chinese Academy of Fishery Sciences (ID Number: YFI 2022-mengyan-201).

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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Data availability

Data will be made available on request.

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